Installment Reconciliation: Business Logic and Technical Reference

Purpose

For every loan installment, the reconciliation checks whether each recorded **settlement** (installment or penalty marked as paid) is backed by **money actually received** (Electronic, Manual, Unblock). The result is one row per installment in `public.installment_reconcilation`, shown in the Metabase **Installment Reconciliation Dashboard**.

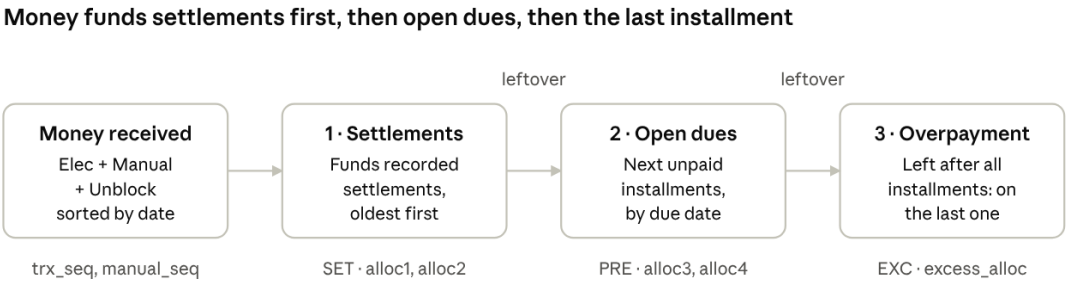
Business rules

Scope

- Loans with status **9** (قائم, active) or **50** (عادي سداد مغلق, closed), products with `DEBT_TYPE < 500`.
- **Early repayment excluded:** any loan that ever had `FLAG = 13` does not appear at all, neither its installments nor its money. It still takes part in matching, so its money does not spill onto the customer's other loans.

How money is matched

1. **Settlements first:** money pays the recorded settlements, oldest first. On the same day, the penalty is paid before its installment.
2. **Then open installments:** money left over goes to the next installments that still have a balance, by due date.
3. **Then overpayment:** money left after every installment of the loan is placed on the loan's last installment.



allocation flow · 3 stages

Manual payments without a loan number are matched at customer level: they go to whichever of the customer's loans has a settlement or installment still waiting for money, oldest first. This prevents payments for an old loan landing on a renewal loan opened the same day.

Statuses

Each installment gets one `RECON_STATUS`, checked in this order (first match wins). Examples use an installment of 5,000 due on 5 Sep.

Status	Meaning	Example	Action
`Not due yet`	The due date has not arrived. Nothing to judge yet	Today is 1 Sep; any payment received sits on it until it is due	None
`Not settled yet`	Due date passed; no settlement and no money	5 Sep passed, the customer paid nothing, the system shows it unpaid	Collections follow-up
`Settled without transaction`	Marked as paid, but no money behind it	Settled 5,000 on 5 Sep; no payment exists for it	**Investigate**
`Partially funded`	Settled for more than the money received	Settled 5,000; only 3,000 came in. `DIFF = -2,000`	**Investigate**
`Overpaid`	Money above every installment of the loan; placed on the last installment	Loan fully settled; the customer paid 150 more. `DIFF = +150`	Refund, or confirm it is a fee
`Paid, not settled`	Money came in for this installment, but it was not settled	Customer paid 5,000 on 5 Sep; the installment is still open in the system	**Settle it**
`Settled before payment`	Amount matches, but the settlement was recorded before the money arrived	Settled on 5 Sep; the 5,000 arrived on 12 Sep	Check the procedure
`Matched`	Settlement fully funded, on time	Paid 5,000 on 5 Sep, settled 5,000 on 5 Sep	None

Exceptions used on the dashboard: `Settled without transaction`, `Partially funded`, `Paid, not settled`, `Overpaid`.

Rare extra row: `Transaction - loan has no installments` is money on a loan with no installment schedule. It has no installment columns.

Key columns

Column	Formula / meaning
`TOTAL_DUE`	Installment value + penalty due (waived penalties excluded)
`TOTAL_SETTLED`	Installment settlement + penalty paid
`REMAINING`	`TOTAL_DUE − TOTAL_SETTLED`: still owed by the customer
`ELEC_AMT`, `MANUAL_AMT`, `UNBLK_AMT`	Money on this installment by transaction type
`FUNDED_AMT`	Sum of the three transaction types
`DIFF`	`FUNDED_AMT − TOTAL_SETTLED`. Positive: more money than settled. Negative: settled without enough money
`FILL_METHOD`	`By customer` when funded by a manual payment that had no loan number
`SORT_DATE`	Due date; used by the dashboard date filter